KPM-Bridge: An Uncertainty-Aware Cross-Implementation Telemetry Fabric for Portable AI xApps in Open 6G RAN

Yassir AL-Karawi and Hamed Al-Raweshidy

***Abstract*—Open Radio Access Network (O-RAN) implementations can emit syntactically valid E2 Service Model for Key Performance Measurement (E2SM-KPM) reports that denote different statistical objects for the same artificial-intelligence (AI) control application, or xApp. Units, entity scope, observation windows, clocks, counter resets, missingness, and hidden filtering may differ silently. KPM-Bridge separates E2 interoperability from semantic and statistical portability by compiling typed canonical contracts, aligning event-time observations, learning an anchor-assisted residual map, and attaching a 48-byte evidence certificate; split-conformal sets cover the canonical vector and a registered xApp functional, so a fixed xApp acts or abstains. On 201,912 public ColO-RAN samples with experiment-level separation, seven baselines, ablations, and controlled profiles, KPM-Bridge reaches 0.694 normalized root-mean-square error and 89.34% canonical-decision agreement across three stationary profiles—matching the strongest portable baselines within bootstrap interval widths, so the contribution is the evidence layer, not raw accuracy. Under the admission gate, admitted actions show 0.209% conditional canonical-action disagreement at 54.45% acceptance, with 94.32% joint coverage. Under severe drift, the monitor detects 88.7% of shifted traces; fail-closed invalidation cuts post-shift disagreement exposure from 86.0% to 4.57% by reducing acceptance to 5.99%; conditional disagreement remains 76.23%. The profiles are controlled stressors, not commercial-vendor measurements.**

***Index Terms*—6G, Open RAN, E2SM-KPM, xApp portability, conformal prediction.**

I. Introduction

OPEN Radio Access Network (O-RAN) architectures disaggregate the radio access network (RAN), expose standardized interfaces, and place programmable applications in a near-real-time RAN Intelligent Controller (Near-RT RIC) [1]–[3]. This direction is central to an artificial-intelligence (AI)-native 6G RAN because a control application, or xApp, can observe several E2 nodes and adapt radio behavior without modifying each baseband stack [4]–[6]. The E2 Service Model for Key Performance Measurement (E2SM-KPM) structures KPM subscription and reporting across the E2 interface [7].[1] Protocol conformance does not, by itself, guarantee that independently implemented reports are interchangeable inputs to a learned model.

Consider an xApp trained on per-user-equipment downlink throughput averaged over 250 ms. Another implementation can expose a cumulative byte counter at cell scope, reset it after a restart, and report every second. Both messages may decode correctly, and a software adapter may label both values "throughput." Yet differentiation, scope disaggregation, event-time alignment, and information loss remain unresolved, and the failure is silent: the decoder succeeds, the xApp produces a number, and the action falls outside the semantics the model was trained on. KPM granularity is itself now an optimization variable in O-RAN digital-twin workflows [8], making explicit treatment of window and freshness more important still.

Open experimentation platforms make AI xApps reproducible [9]–[13]. xDevSM recently showed that application logic can be insulated from low-level service-model procedures and run across heterogeneous open-source RAN stacks [7], [14]. KPM-Bridge addresses the next boundary: protocol and application-programming-interface (API) portability answer whether an xApp can request and decode a measurement, whereas semantic portability asks whether that measurement denotes the expected quantity, scope, time support, and statistical regime.

The central design rule is that a canonical KPM vector is never emitted alone. It is paired with evidence specifying what was mapped, how old and complete it is, how its uncertainty was calibrated, and whether the calibration regime remains valid. A fixed xApp may then act or abstain under an explicit quality budget rather than treating every decoded report as equally trustworthy.

This paper makes five contributions.

- It defines a machine-checkable KPM contract over quantity, physical unit, entity scope, aggregation, window, clock, counter/reset semantics, schema version, and provenance. A minimum-cost compiler fails closed when a canonical target has no compatible source.
- It introduces an event-time transport model and an anchor-assisted mapper whose temporal features cover lag, smoothing, missingness, and resetting counters while preserving type constraints.
- It combines a joint split-conformal telemetry set with an xApp-specific functional interval, a selective decision gate, and a calibrated multivariate anchor-residual drift monitor.
- It proves type soundness, a translation-error decomposition, finite-sample coverage under explicit exchangeability,

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[1]Normative E2SM-KPM releases are distributed through the official O-RAN specification portal (accessed July 13, 2026); O-RAN specifications do not carry Crossref DOIs.

downstream decision stability, and online complexity bounds.
- It provides a deterministic trace-backed evaluation with seven portable baselines, two nonportable references, six ablated variants, cluster-bootstrap intervals, and sensitivity to anchors, loss, lag, miscoverage, and drift.

The contribution is Open-RAN-specific: contracts are keyed by E2 node, service-model revision, measurement identifier, and xApp model across independently implemented producers. KPM-Bridge certifies measurement compatibility and decision stability under stated assumptions; it neither proves that an xApp objective is safe nor authenticates compromised code. Execution integrity and interface security remain orthogonal [15].

II. Related Work

O-RAN service-model implementations expose KPM subscription, indication, and measurement information structures through the E2 interface [3], [7]. General network-telemetry guidance separates data acquisition, export, analysis, and use [16], [17], while the Network Management Datastore Architecture emphasizes explicit operational state and provenance [18]. KPM-Bridge specializes these principles for learned Near-RT RIC inputs: it treats scope, time support, counter semantics, and calibration state as part of the feature type.

ColO-RAN demonstrated closed-loop machine-learning xApps on Colosseum and released traces spanning seven base stations, 42 user equipment (UE) devices, three traffic slices, and three schedulers [9]. OpenRAN Gym added repeatable data-collection and testing workflows [10], [11], and FlexRIC, OrchestRAN, Open AI Cellular (OAIC), and ns-O-RAN broadened software-development, orchestration, testbed, and simulation access [12], [13], [19], [20]. These platforms are the evidence backbone used here, but their objective is experimentation or control orchestration, not a cross-implementation uncertainty contract.

Recent application and architecture studies expose why this boundary matters. A native-AI digital-twin architecture connects E2/A1/O1 services to shadow testing, guarded control, and rollback [21]. Such systems are direct telemetry consumers and deployment stressors, not semantic-interoperability mechanisms: they do not define a typed cross-implementation KPM contract, a finite-sample translation set, or drift-aware abstention.

Open-RAN intelligence frameworks establish the role of data-driven xApps and coordinated model placement [4], [5], [20]. FlexRIC provides a portable software-development kit (SDK) at the RIC boundary [19], whereas xDevSM hides service-model mechanics behind developer-facing APIs [7]. Its 2026 extension unifies observability and control across heterogeneous OpenAirInterface (OAI)- and srsRAN-based deployments and reports stack-specific KPM granularity and scheduler behavior [14]. It is the closest complementary work: KPM-Bridge neither disputes its portability claim nor replaces its APIs, but can sit behind such an API to decide whether two decoded metrics are semantically and statistically fit for the same trained input.

Table I separates protocol or API portability from the semantic, temporal, and statistical evidence required by a fixed learned input. The comparison also makes the boundary of the contribution explicit: KPM-Bridge complements E2 decoders and portable xApp APIs rather than replacing them.

Domain-adaptation theory bounds target error using source error and distribution divergence, but also retains a joint-labeling term that cannot be removed by marginal alignment [22]. Correlation alignment (CORAL), adversarial adaptation, and optimal transport (OT) are established ways to reduce distribution mismatch [23]–[26]. They do not determine whether two columns represent the same physical quantity or entity; an unconstrained map can align cell load to per-UE throughput while matching marginals. KPM-Bridge therefore applies statistical correction only after type-compatible matching.

Conformal prediction converts held-out nonconformity scores into finite-sample prediction sets under exchangeability [27], [28]; weighted methods extend the framework beyond exchangeability under quantified departures [29]. Reject-option and set-valued classification formalize the error–coverage tradeoff under abstention [30], [31], and concept-drift methods detect and adapt to invalid operating regimes [32]–[36]. Our contribution is their joint placement at the O-RAN telemetry boundary, with separate certificates for the canonical vector and the downstream xApp functional.

III. Problem Formulation

Let $\mathcal{V}$ denote the connected RAN implementations and let $z_t \in \mathcal{Z} \subseteq \mathbb{R}^d$ be the canonical KPM state at event time t. In the released benchmark $d = 8$, but the formulation is general. Multi-metric controllers can depend jointly on queue, energy, latency, signal-to-interference-plus-noise ratio, and fairness observations, as illustrated by energy-aware off-grid O-RAN control [37]; compatibility is therefore feature- and scope-specific. A raw stream i from implementation v has contract

$$c_{v,i} = (q_{v,i}, u_{v,i}, e_{v,i}, a_{v,i}, w_{v,i}, \tau_{v,i}, \rho_{v,i}, \nu_{v,i}), \tag{1}$$

where q is quantity, u unit, e entity scope, a aggregation, w observation window, τ clock basis, ρ gauge/delta/cumulative and reset semantics, and ν provenance plus schema version. A canonical feature k has target contract $c_k^\star$.

Definition 1 (Type-compatible candidate)**.** *A source $c_{v,i}$ may supply $c_k^\star$ only if quantity, physical dimension, and entity scope agree. A unit difference creates an explicit conversion. Differences in aggregation, window, clock, or counter semantics create required transforms and residual uncertainty, not implicit equivalence.*

Let $m_{v,t} \in \{0,1\}^{p_v}$ be the observation mask. The report vector is modeled as

$$y_{v,t} = m_{v,t} \odot \left[g_v(\Phi_v(z_{\leq t})) + \epsilon_{v,t}\right], \tag{2}$$

where $\odot$ denotes elementwise multiplication, g_v contains declared unit/counter operations plus an identified residual map, Φ_v contains windowing and delay, and $\epsilon_{v,t}$ collects noise and quantization. This model does not assume that g_v is globally invertible. Aggregation and missingness can destroy information.

TABLE I

COMPARISON OF REPRESENTATIVE O-RAN TELEMETRY AND PORTABILITY WORK BY INTERFACE SCOPE, KPM SEMANTICS, TEMPORAL REPAIR, UNCERTAINTY SETS, AND DRIFT HANDLING.

Work	Primary layer	Cross-stack E2/API	Typed KPM meaning	Temporal/counter repair	Finite-sample set	Drift-aware abstention
O-RAN/E2SM studies [3], [7]	architecture/service model	standardized protocol	measurement catalogue	report semantics	no	no
ColO-RAN/OpenRAN Gym [9], [11]	experimentation	platform adapters	dataset-specific	pipeline-specific	no	no
ns-O-RAN [12]	RIC–simulation integration	yes	scenario-specific	simulator clock	no	no
FlexRIC/OrchestRAN [19], [20]	SDK/orchestration	yes	model-defined	runtime-specific	no	no
xDevSM [7], [14]	developer API	yes	unified metric access	service-model handling	no	no
KPM granularity xApp [8]	digital-twin reporting	xApp-level	granularity objective	adaptive granularity	no	no
KPM-Bridge	AI-input evidence layer	complementary	quantity/scope/unit contract	explicit event time and reset	joint and functional	yes

KPM-Bridge returns

$$B_v(y_{v,\le t}) = (\hat{z}_t, \mathcal{U}_t, \gamma_t), \tag{3}$$

where $\mathcal{U}_t$ is a calibrated set and γ_t contains schema hash, mapping identifier, calibration epoch, support, age, radius, and drift flags. Its point-estimate component is denoted by $M_v(y_{v,\le t}) = \hat{z}_t$. A fixed xApp with scalar decision functional $\ell(z)$ and threshold θ acts only if the certified interval for $\ell(z_t)$ lies wholly on one side of θ.

The mapper is fitted from a limited paired-anchor set $\mathcal{D}_v^A = \{(y_{v,t}, z_t)\}$ and an independent calibration set $\mathcal{D}_v^C$. For the positive scale matrix $D = \mathrm{diag}(s_1, \ldots, s_d)$, with $s_k > 0$ for every canonical feature, the statistical objective is

$$\min_{M_v} \sum_{(y,z)\in\mathcal{D}_v^A} \left\| D^{-1}(M_v(y) - z) \right\|_2^2 + \lambda\Omega(M_v), \tag{4}$$

subject to a contract-compiled input plan. Here $\lambda \ge 0$ controls regularization and $\Omega(M_v)$ penalizes mapper complexity. In the benchmark, s_k is the larger of the fit-prefix sample standard deviation and interquartile range divided by 1.349, with a unit fallback only for a degenerate feature. Thus D and all regularization choices are estimated from training data only. Equation (4) separates statistical fitting from the semantic feasibility constraints enforced before fitting. The operational objective is not forced action coverage. It is low canonical-action disagreement among admitted actions while exposing abstention.

The considered shifts include declared unit and counter changes, window/lag, noise, quantization, entry loss, hidden monotone filtering, and post-deployment mean shift. We assume the E2 termination authenticates message origin. Malicious function declarations, compromised xApp execution, and adversarial anchor corruption are outside scope; the certificate's provenance fields are hooks for those mechanisms, not substitutes for them.

IV. ARCHITECTURE AND ALGORITHMS

Fig. 1 places KPM-Bridge between E2 decoding and the portable xApp feature interface. The data path remains local to the Near-RT RIC: it does not add a new radio control round trip. The evidence plane changes more slowly and stores versioned contracts, anchors, calibration scores, and drift state.

A. Type-Safe Contract Compilation

Let $C_v = \{c_{v,i}\}_{i=1}^{p_v}$ and $C^\star = \{c_k^\star\}_{k=1}^{d}$ denote the source and canonical contract sets, respectively. The compiler builds a bipartite graph between these sets. Incompatible edges have infinite cost. A compatible edge has

Input: authenticated registry records C_v, targets $C^\star$, per-target budgets, version ν

1. Validate registry binding, mandatory fields, provenance, and schema versions.
2. For each (i, k), remove quantity, dimension, or scope mismatches; otherwise compute κ_{ik} by (5).
3. Solve the minimum-cost injective assignment over finite edges.
4. If any target is unmatched or exceeds its budget, return UNSUPPORTED with a reason code.
5. Emit target-ordered unit/counter transforms and clock/window obligations.
6. Compute one Secure Hash Algorithm 256 (SHA-256) digest over the canonical serialization of $(C_v, C^\star, \text{plan}, \nu)$ and its semantic budget; bind bytes 0–15 as the schema hash and bytes 16–23 as the mapping identifier.

Output: executable transform plan, or a fail-closed reason code

Algorithm 1: Compilation of source KPM contracts into a type-safe canonical transform plan.

$$\begin{aligned}\kappa_{ik} = {}& \beta_a \mathbf{1}[a_i \ne a_k^\star] + \beta_w \left| \log \frac{w_i}{w_k^\star} \right|_{[0,K_w]} \\ &+ \beta_\tau \mathbf{1}[\tau_i \ne \tau_k^\star] + \beta_\rho \mathbf{1}[\rho_i \ne \rho_k^\star].\end{aligned} \tag{5}$$

where the nonnegative coefficients $\beta_a, \beta_w, \beta_\tau$, and β_ρ are operator-set semantic penalties. The observation windows satisfy $w_i, w_k^\star > 0$, and $|x|_{[0,K_w]} = \min\{|x|, K_w\}$, with $K_w > 0$, prevents a single window mismatch from dominating the assignment. The minimum-cost injective assignment is solved after type pruning. Each selected edge emits deterministic unit and counter transforms, reset handling, clock/window obligations, and provenance binding. A window mismatch is not assumed invertible; it enters the temporal design and residual uncertainty. An absent target or a cost above its budget fails closed.

Fig. 2 illustrates the two-stage rule encoded by (5) and Algorithm 1: quantity, dimension, and scope are hard type constraints, whereas unit, counter, clock, and window differences generate explicit transforms and uncertainty obligations.

The schema hash and mapping identifier occupy disjoint fields of one SHA-256 digest, matching the reference implementation and the fixed widths in Table II.

B. Event-Time Alignment and Residual Mapping

The declared plan first converts units and counters. A cumulative counter b_t with sampling interval Δ_t is mapped to $(b_t - b_{t-1})/\Delta_t$. A negative delta raises a reset flag: b_t/Δ_t is used only when reset metadata confines the event to the sampling boundary, as in the controlled profiles; otherwise the derived rate is missing. Missing adjacent values are never extrapolated.

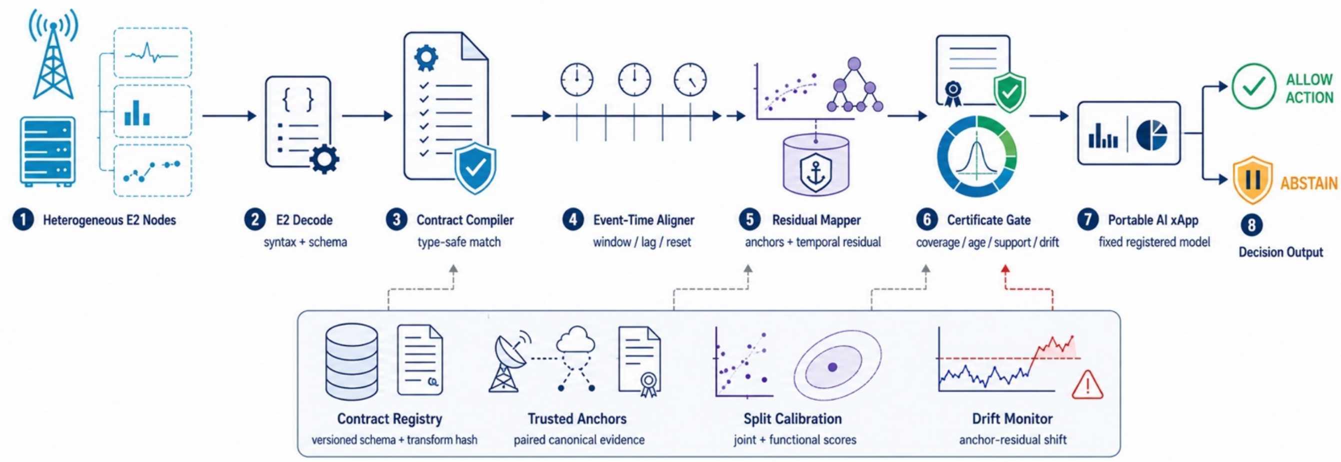

Fig. 1. KPM-Bridge placement between E2 decoding and a portable AI xApp. The data path compiles and aligns reports; the evidence plane supplies versioned contracts, trusted anchors, calibration, and drift state.

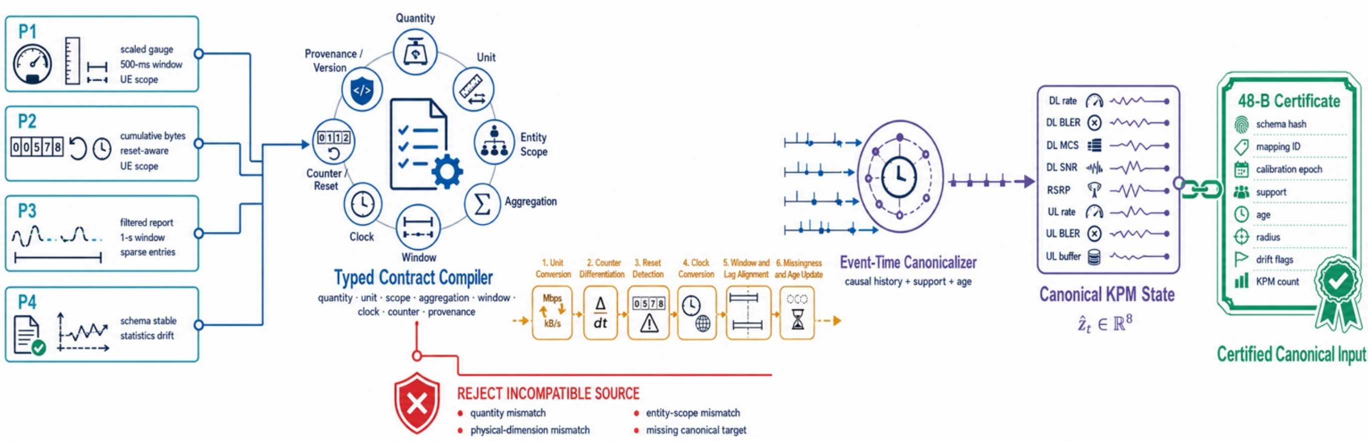

Fig. 2. Typed canonicalization from heterogeneous KPM profiles to a certified eight-dimensional state. Incompatible quantities or scopes fail closed; compatible streams receive unit/counter transforms and explicit clock, window, and missingness obligations.

For each canonical instant, the reference implementation forms a causal design vector

$$x_t = [\tilde{y}_t, \tilde{y}_{t-1}, \tilde{y}_{t-2}, \tilde{y}_{t-4}, \tilde{y}_{t-8}, \Delta\tilde{y}_t, m_t, \mathrm{age}_t, \mathrm{support}_t], \tag{6}$$

where $\tilde{y}$ is the contract-normalized report, $\Delta\tilde{y}_t = \tilde{y}_t - \tilde{y}_{t-1}$ is the causal first difference, m_t is the observation mask, and age and support summarize event-time staleness and completeness. One histogram gradient-boosted regressor per canonical KPM predicts the standardized target [38]. The released configuration uses 70 trees, at most 15 leaves, depth five, and deterministic seeds. The ablation includes a temporal ridge map.

C. Joint and Functional Calibration

On held-out calibration samples, the joint score is

$$R_j = \left\| D^{-1}(\hat{z}_j - z_j) \right\|_2. \tag{7}$$

For n_c scores and $\alpha \in (0,1)$, let

$$\begin{aligned} \hat{q}_\alpha &= R_{(\lceil (n_c+1)(1-\alpha) \rceil)}, \\ \mathcal{U}_t &= \{z : \left\| D^{-1}(z - \hat{z}_t) \right\|_2 \le \hat{q}_\alpha\}. \end{aligned} \tag{8}$$

The order-statistic convention is $R_{(n_c+1)} = +\infty$, which preserves validity when the requested α is smaller than $1/(n_c+1)$. This is the standard split-conformal quantile construction under exchangeability [27], [28]. The generic ball can be conservative for a particular xApp. KPM-Bridge therefore also calibrates

$$\begin{aligned} S_j &= |\ell(\hat{z}_j) - \ell(z_j)|, \\ \hat{q}^{\ell}_\alpha &= S_{(\lceil (n_c+1)(1-\alpha) \rceil)}, \\ I^{\ell}_t &= [\ell(\hat{z}_t) - \hat{q}^{\ell}_\alpha, \ell(\hat{z}_t) + \hat{q}^{\ell}_\alpha]. \end{aligned} \tag{9}$$

The functional interval does not replace $\mathcal{U}_t$; it is tied to the registered model hash, and a model change invalidates it. The authenticated registry uses the mapping identifier and calibration epoch to resolve that model hash, $\hat{q}^{\ell}_\alpha$, θ, and the applicable quality budgets. Thus the compact sidecar need not duplicate the fixed functional quantile. The two coverages are marginal; simultaneous coverage would require a multiplicity adjustment or one joint score.

Drift invalidation and admission: Trusted anchors periodically reveal standardized residual vectors $r_t = D^{-1}(\hat{z}_t - z_t)$. From calibration anchors, KPM-Bridge estimates (μ_r, Σ_r).

Input: report/metadata, compiled plan, mapper, calibration epoch, quality budgets

1 Verify schema hash, mapping identifier, source provenance, and epoch.
2 Apply unit/counter transforms, enforce clock/window obligations, and update (6).
3 Predict $\hat{z}_t$ and attach $\hat{q}_\alpha$, support, age, and calibration epoch.
4 At an anchor, update H_t by (10); invalidate the epoch on a trigger.
5 Compute I_t^ℓ by (9) and evaluate support, age, drift, and threshold crossing.
6 Serialize the 48-byte certificate; emit the xApp action if admitted, else ABSTAIN with a reason code.

Output: canonical vector, certificate, action/abstention, and reason code

Algorithm 2: Certified streaming inference with event-time alignment, calibrated uncertainty, drift checks, and selective action.

With $W = 5$ anchors, the monitor computes

$$H_t = \sqrt{W(\bar{r}_t - \mu_r)^\top (\Sigma_r + 0.1I)^{-1}(\bar{r}_t - \mu_r)}. \tag{10}$$

where $\bar{r}_t = W^{-1}\sum_{j=0}^{W-1} r_{t-j}$ and the $d \times d$ identity I makes $0.1I$ a covariance regularizer. Anchors arrive every ten samples (2.5 s in this corpus). The threshold is the 97.5th percentile of the *per-stream maximum* calibration score, which addresses repeated testing more directly than a pointwise quantile. Candidate windows are ranked without test data on held-out `exp1` streams, subject to a predeclared 5% validation false-alarm ceiling and synthetic 1.25-scale shifts; this procedure selects $W = 5$. A trigger invalidates the calibration epoch rather than widening a stale interval. Recalibration is a separate control-plane action.

Admission gate and selective metrics: An event is *admitted*, $g_t = 1$, only when support and age pass, the calibration epoch is drift-valid, and the functional interval avoids the threshold. Write $g_t = g_t^{\text{sup}} g_t^{\text{age}} g_t^{\text{epoch}} g_t^{\text{func}}$, where $g_t^{\text{sup}} = \mathbf{1}\{\text{support}_t \geq s_{\min}\}$, $g_t^{\text{age}} = \mathbf{1}\{\text{age}_t \leq a_{\max}\}$, $g_t^{\text{epoch}} = \mathbf{1}\{\text{epoch valid}\}$, and $g_t^{\text{func}} = \mathbf{1}\{\theta \notin I_t^\ell\}$. Then $g_t = 1$ emits $a(\hat{z}_t)$ and $g_t = 0$ abstains with a reason code. Over T events with $\mathcal{I}_{\text{adm}} = \{t : g_t = 1\}$, the acceptance rate, selective (admission-conditional) error, and joint disagreement exposure are

$$\text{acc} = \frac{1}{T}\sum_t g_t, \qquad \text{serr} = \frac{1}{|\mathcal{I}_{\text{adm}}|}\sum_{t\in\mathcal{I}_{\text{adm}}} \mathbf{1}\{a(\hat{z}_t) \neq a(z_t)\},$$
$$\text{exp} = \frac{1}{T}\sum_t g_t\,\mathbf{1}\{a(\hat{z}_t) \neq a(z_t)\} = \text{acc}\cdot\text{serr}. \tag{11}$$

Thus $\text{serr} = \Pr\{a(\hat{z}_t) \neq a(z_t) \mid g_t = 1\}$ is measured directly, whereas Proposition 5 bounds only the joint $\Pr\{\theta \notin I_t^\ell, a(\hat{z}_t) \neq a(z_t)\}$; the identity $\text{exp} = \text{acc}\cdot\text{serr}$ is instantiated post-shift by $5.99\% \times 76.23\% = 4.57\%$.

Algorithm 2 is the online counterpart of the compiler. It binds the temporal design in (6), the calibrated functional interval in (9), and the drift statistic in (10) to one fail-closed admission decision.

The encoded sidecar is fixed at 48 B: 16-byte schema hash; two 8-byte identifiers for mapping and calibration epoch; three 4-byte floats for support, age, and the generic radius; and two 2-byte fields for flags and KPM count. The functional quantile is resolved from the authenticated registry as described above, and the feature vector is carried separately.

TABLE II
FIXED 48-BYTE CERTIFICATE LAYOUT AND THE OPERATIONAL MEANING OF EACH FIELD.

Field	Bytes	Purpose
Schema hash	16	binds source, target, and plan
Mapping identifier	8	selects fitted mapper
Calibration epoch	8	invalidation boundary
Support	4	observed KPM fraction
Age	4	event-time staleness in ms
Radius	4	generic conformal radius
Flags	2	drift/schema/quality bits
KPM count	2	canonical dimension guard
Total	48	network byte order

D. O-RAN Placement and Certificate Lifecycle

KPM-Bridge is an internal evidence layer, not a replacement for the E2 Application Protocol (E2AP) or E2SM-KPM. The E2 termination handles subscription and Abstract Syntax Notation One decoding, then passes values and report metadata to a local bridge adapter, which retrieves a contract from the E2 node identity, service-model revision, measurement identifier, and operator policy. A deployment may provision that registry from the Service Management and Orchestration framework or the non-real-time RIC (Non-RT RIC); no new standardized A1, O1, or R1 message is claimed. The xApp-facing API returns a canonical vector, certificate, and reason code.

Fig. 3 separates five persistent states. UNSEEN means that no signed contract is known. COMPILED means that type matching and transform planning succeeded, but no statistical guarantee exists. CALIBRATED binds the mapper, scale matrix, generic score, functional score, and drift statistics to an epoch. ACTIVE permits evaluation of the quality gate. INVALID is fail-closed and preserves a reason; it is not automatically converted back to ACTIVE. A schema change returns to compilation, whereas a pure statistical alarm requires new anchors and calibration.

Certificate serialization uses network byte order and fixed-width fields. The 16-byte hash is a truncated cryptographic digest for compact binding, not a digital signature. The surrounding RIC platform must authenticate registry updates. The mapping and epoch identifiers prevent a consumer from combining a vector from one model with intervals from another. Floating-point support, age, and radius are sufficient for the reference implementation; a safety-critical deployment may choose fixed point and a versioned layout.

Table II accounts for every serialized byte; the certificate is a local evidence sidecar, while the canonical feature vector remains a separate payload.

The reason codes distinguish "unsupported" from "uncertain." An unsupported type should not be fixed by collecting more samples. An uncertain but compatible stream may recover after a fresh report or additional anchor. This distinction is useful for orchestration because it prevents repeated refinement requests for a fundamentally incompatible KPM. Table III maps each invalidation condition to an immediate fail-closed response and a distinct recovery path.

V. ANALYTICAL GUARANTEES

Proposition 1 (Compiler type soundness)**.** *If Algorithm 1 returns a mapping, every selected source and target have equal quantity, physical dimension, and entity scope.*

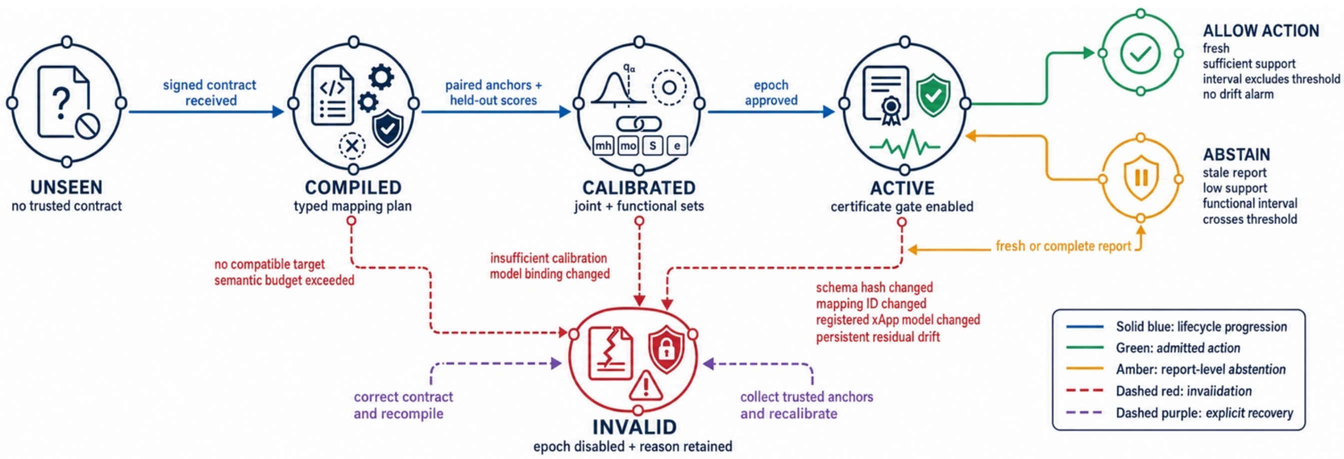

Fig. 3. Certificate lifecycle from an unseen contract to an active or invalid state. Schema or model changes force recompilation, whereas statistical drift requires new trusted anchors and recalibration.

TABLE III
FAIL-CLOSED CONDITIONS, EMITTED REASON CODES, AND REQUIRED RECOVERY ACTIONS.

Condition	Observable evidence	Immediate action	Recovery
Quantity/dimension/scope mismatch	no finite compiler edge	reject mapping	correct contract or source KPM
Unknown reset or adjacent counter loss	invalid delta and reduced support	mark feature missing	next valid delta or anchor
Schema/model hash change	certificate binding mismatch	invalidate epoch	recompile and recalibrate
Age or support budget violation	$\mathrm{age}_t > a_{\max}$ or $\mathrm{support}_t < s_{\min}$	abstain with reason	fresh/complete report
Functional interval crosses threshold	$\theta \in I_t^\ell$	abstain	refine, wait, or operator policy
Anchor-residual mean shift	H_t exceeds stream threshold	invalidate epoch	diagnose and recalibrate

Proof: An edge is inserted only after all three equality predicates pass. The assignment solver selects only finite inserted edges. Therefore every returned pair satisfies the predicates. Unit, temporal, and counter differences may remain, but they are explicit transforms rather than silent type substitutions. ■

Type soundness does not imply identifiability. A cell mean cannot generally recover per-UE values even when both use a throughput unit; the scope predicate rejects that edge. A one-second mean and a 250-ms mean have compatible quantity and scope but unequal information. Their edge can be admitted only with a window transform and uncertainty.

Proposition 2 (Translation-error decomposition)**.** *Unless a subscript states otherwise, the norms in this proposition are Euclidean. Let u_t^0 be the noise-free, full-support output of the declared transform; u_t^m its masked version; and u_t^ϵ the noisy masked input. Let F_v^0 be the fitted calibration-regime realization of M_v, and let $\bar{z}_t^0$ and $\bar{z}_t$ be the information-preserving targets under the calibration and current regimes. Then $\hat{z}_t = F_v^0(u_t^\epsilon)$ obeys*

$$\begin{aligned}\|\hat{z}_t - z_t\| \le E_{\mathrm{temp}}(t) + E_{\mathrm{model}}(t) + E_{\mathrm{noise}}(t)\\ + E_{\mathrm{miss}}(t) + E_{\mathrm{drift}}(t),\end{aligned} \tag{12}$$

where $E_{\mathrm{noise}} = \left\|F_v^0(u_t^\epsilon) - F_v^0(u_t^m)\right\|$, $E_{\mathrm{miss}} = \left\|F_v^0(u_t^m) - F_v^0(u_t^0)\right\|$, $E_{\mathrm{model}} = \left\|F_v^0(u_t^0) - \bar{z}_t^0\right\|$, $E_{\mathrm{drift}} = \left\|\bar{z}_t^0 - \bar{z}_t\right\|$, *and* $E_{\mathrm{temp}} = \|\bar{z}_t - z_t\|$.

Proof: Insert $F_v^0(u_t^m)$, $F_v^0(u_t^0)$, $\bar{z}_t^0$, and $\bar{z}_t$ between $\hat{z}_t$ and z_t, then apply the triangle inequality. Compiler soundness removes a dimension/scope substitution term but not temporal or information-loss terms. ■

Proposition 3 (Non-identifiability under aggregation)**.** *Suppose a source reports only $y = Az$, where A is a linear aggregation operator, and there exist $z^{(1)} \neq z^{(2)}$ in the admissible state space with $Az^{(1)} = Az^{(2)}$. For any deterministic point mapper $M(y)$, at least one of the two states has*

$$\left\|M(y) - z^{(j)}\right\|_2 \ge \frac{1}{2}\left\|z^{(1)} - z^{(2)}\right\|_2. \tag{13}$$

Proof: Both states generate the same report and therefore the same estimate. If both errors were smaller than half their separation, the triangle inequality would imply $\left\|z^{(1)} - z^{(2)}\right\|_2 < \left\|z^{(1)} - z^{(2)}\right\|_2$, a contradiction. ■

This proposition formalizes why the bridge cannot manufacture per-UE state from a cell aggregate. Anchors or a structural prior can select among otherwise indistinguishable states, but the resulting dependence must appear in the mapping identifier and uncertainty. The compiler's scope check rejects the most direct form of this failure before fitting.

Assumption 1 (Calibration exchangeability)**.** *Conditional on a fixed schema, mapping, model, and non-drift regime, the test score and the n_c calibration scores are exchangeable. Calibration data are not used to fit the mapper.*

Proposition 4 (Finite-sample coverage)**.** *Under calibration exchangeability, the set in (8) satisfies*

$$\Pr\{z_{n_c+1} \in \mathcal{U}_{n_c+1}\} \ge 1 - \alpha. \tag{14}$$

The functional interval in (9) obeys the analogous inequality for $\ell(z)$.

Proof: The rank of the test nonconformity score among $n_c + 1$ exchangeable scores is uniform up to ties. The order statistic in (8) excludes at most an α fraction of ranks. Applying the same argument to S_j gives functional coverage [27], [28]. ■

Temporal traces are not literally independent. The experiment separates repetitions and subsamples calibration points, but the formal statement remains conditional on exchangeability. Section VII reports empirical coverage rather than relabeling it as a theorem under arbitrary autocorrelation.

Proposition 5 (Decision stability)**.** *For a threshold action* $a(z) = \mathbf{1}[\ell(z) \geq \theta]$, *if* $\theta \notin I_t^\ell$ *and* $\ell(z_t) \in I_t^\ell$, *then* $a(\hat{z}_t) = a(z_t)$. *Under calibration exchangeability, the joint probability of passing the interval gate and disagreeing with the canonical action is at most* α*; no conditional selective-error bound is claimed.*

Proof: When I_t^ℓ lies strictly above or below θ, all contained values induce the same action. Functional coverage gives $\Pr\{\theta \notin I_t^\ell,\, a(\hat{z}_t) \neq a(z_t)\} \leq \alpha$. Support, age, and drift filters preserve this joint bound because they only remove events. They do not imply the conditional bound $\Pr\{a(\hat{z}_t) \neq a(z_t) \mid \text{admit}\} \leq \alpha$; selective error is therefore measured separately. ■

For a general utility $J(a, z)$ and action set $\mathcal{A}$, the xApp may instead use robust optimization

$$a_t^\star \in \arg\max_{a \in \mathcal{A}} \min_{z \in \mathcal{U}_t} J(a, z), \tag{15}$$

which follows the established robustness principle [39]. The evaluated binary risk xApp uses the functional interval because it is directly aligned with its registered threshold rule.

Computationally, let $n_{\text{asn}} = p_v + d$ be the padded assignment dimension, h the temporal design width, N_{tr} the trees per output, L_{tr} their maximum depth, $N_{\text{leaf}} \leq 2^{L_{\text{tr}}}$ their maximum leaves, and W the drift-window length. Dense assignment is $O(n_{\text{asn}}^3)$ time and $O(p_v d)$ memory, but it runs on schema change, not per report. Contract inversion and feature updates are $O(hd)$; tree inference is $O(dN_{\text{tr}}L_{\text{tr}})$; set construction is $O(d)$; and the multivariate anchor score is $O(d^2 + Wd)$ at anchor epochs. Model storage is $O(dN_{\text{tr}}N_{\text{leaf}})$. These bounds are independent of calibration-set size after quantiles and drift statistics are materialized.

VI. Experimental Design

A. Trace Backbone and Controlled Profiles

The benchmark pins commit `bd86629` of the public ColO-RAN dataset [9]. Its Rome-static-medium scenario has seven base stations (BSs), 42 UEs, 10 MHz bandwidth, 50 physical resource blocks (PRBs), and enhanced mobile broadband (eMBB), machine-type communication (MTC), and ultra-reliable low-latency communication (URLLC) slices. Round-robin, water-filling, and proportional-fair schedulers are available across 28 resource allocations.

Our deterministic subset crosses three schedulers, training configurations `tr0`, `tr9`, and `tr18`, experiments `exp1`/`exp2`, BS1/BS4, and one UE per traffic class: 108 files and 208,585 raw rows. Two traces with fewer than 400 attached samples are excluded by a predeclared rule. After a four-sample horizon, 53 `exp1` traces (109,539 rows) fit and calibrate the model and 53 independent `exp2` traces (92,373 rows) test it. Files are hash-verified; raw data under the GNU General Public License version 3.0 (GPL-3.0) are downloaded from upstream rather than redistributed.

The canonical vector contains reported downlink (DL) rate, DL block error rate (BLER), DL modulation and coding scheme (MCS), DL signal-to-noise ratio (SNR), reference signal received power (RSRP), uplink (UL) rate, UL BLER, and UL buffer. Names refer to dataset fields; the study does not relabel source units as standardized commercial KPMs.

Controlled implementation profiles: Profiles P1–P4 in Table IV are deterministic transformations of each real trace. They vary scale, unit representation, window, lag, noise relative to training scale, entry loss, burst loss, quantization, hidden monotone filtering, and counter reset. P4 changes at 55% of every test trace using a 1.25-standard-scale persistent offset, a smaller nonlinear gain change, doubled noise, and two additional lag samples. No profile is assigned to a named vendor.

P0, an identity reference with 0.2% noise, is used only for software checks. P1–P3 assess stationary portability. P4 evaluates invalidation under deliberately severe unannounced drift. The random seed is 20,260,712; all profile seeds are stable hashes of profile and trace identifiers.

B. xApp, Baselines, and Metrics

The fixed xApp predicts one-second quality-of-service (QoS) risk from the current canonical vector. For each traffic class, risk is one when the next-four-sample mean downlink rate is below its training-only 35th percentile or its block error rate is above the training-only 75th percentile. The thresholds, feature statistics, and class-balanced logistic model use only the first 60% of each canonical `exp1` trace; the four-sample prediction horizon is removed at the split boundary. Its registered functional is the fitted logit

$$\ell(z) = w^\top[(z - \mu_x) \oslash \sigma_x] + b, \tag{16}$$

where w and b are the logistic coefficients, μ_x and σ_x are fit-prefix feature location and scale, and $\oslash$ denotes elementwise division. On canonical `exp2`, the model's area under the receiver-operating-characteristic curve (AUROC) is 0.755, Brier score 0.205, and risk prevalence 44.9%. These numbers characterize the instrument; KPM-Bridge does not claim to improve its intrinsic model accuracy.

An intervention costs 0.2 and a missed risk costs 1, so the probability threshold is 0.2. Equivalently, the interval gate in (9) uses the logit threshold $\theta = \log(0.2/0.8) \simeq -1.386$ with (16). Regret to a clairvoyant label is 0.8 for a miss and 0.2 for an unnecessary intervention. Portability is primarily measured by agreement with the *same* fixed xApp on canonical input, thereby isolating telemetry shift from source-model error.

Fitting and calibration: For every `exp1` trace, the first 60% is available to the mapper, and 10% of that prefix is selected as evenly spaced paired anchors. The remaining 40% is never used for fitting; every fourth sample supplies conformal calibration. The full `exp2` repetition is untouched test data. Gradient boosting uses fixed hyperparameters across profiles.

Baselines are: (i) direct raw input with median fill; (ii) declared contract conversion alone; (iii) featurewise z-score

TABLE IV
CONTROLLED IMPLEMENTATION PROFILES APPLIED TO PUBLIC COLO-RAN TRACES; PROFILE NAMES DENOTE STRESS SETTINGS RATHER THAN VENDORS.

Profile	Declared representation	Window	Lag	Entry loss	Noise	Quantization	Additional condition
P1	scaled units / gauges	2 samples	1 sample	3%	2%	1%	mild hidden filter
P2	rates as resetting counters	3 samples	1 sample	8%	3%	1.5%	resets every 257 samples
P3	scaled units / gauges	4 samples	2 samples	18%	5%	3%	sparse filtered reports
P4	scaled units / gauges	2 samples	1 sample	5%	2.5%	1.2%	shift at 55% of each test trace

TABLE V
INFORMATION AND SUPERVISION MADE AVAILABLE TO EACH EVALUATED METHOD FAMILY.

Method family	Contracts	Paired anchors	Task labels
Direct / distribution	no	no	no
Contract only	yes	no	no
Anchor/temporal ridge	yes	10%	no
KPM-Bridge	yes	10%	no
Deployment retrain	no	no	60% fit prefix
Oracle canonical	canonical	n/a	source model

alignment; (iv) CORAL [23]; (v) Gaussian OT [25], [26]; (vi) supervised current-sample anchor ridge; and (vii) temporal ridge. We also report deployment-specific logistic retraining on the same 60% fit prefixes (85.58% mean P1–P3 agreement) and an oracle canonical input (100%). Deployment retraining is not portable and has access to about ten times more task labels than KPM-Bridge's mapping anchors.

For clarity, the distribution baselines use the same anchor-budget samples but ignore their eventwise pairing. With raw mean/covariance (μ_y, C_y) and canonical training mean/covariance (μ_z, C_z), z-score alignment is featurewise. CORAL and Gaussian OT use

$$T_{\mathrm{CORAL}}(y) = (y - \mu_y)C_y^{-1/2}C_z^{1/2} + \mu_z,$$
$$T_{\mathrm{OT}}(y) = (y - \mu_y)C_y^{-1/2}(C_y^{1/2}C_zC_y^{1/2})^{1/2}C_y^{-1/2} + \mu_z. \quad (17)$$

Equation (17) uses $10^{-5}I$ covariance regularization. Anchor ridge is multivariate and supervised but sees only the current report. Temporal ridge uses exactly the causal design in (6). Thus the comparison isolates nonlinear residual capacity from temporal context and semantic preprocessing; it does not weaken a baseline by withholding an available history.

Table V prevents two interpretation errors. Distribution alignment is unsupervised and therefore less costly, but it has no eventwise correctness signal. Deployment retraining solves a different problem: it consumes deployment-specific task labels and changes the xApp. Its result is not evidence that the original model is portable.

Canonical error is pooled normalized root-mean-square error (NRMSE), $\sqrt{\mathbb{E}\,\|D^{-1}(\hat{z}-z)\|_2^2/d}$. Other metrics are normalized absolute error, agreement with the fixed xApp on canonical input, label regret, joint coverage, set radius, action acceptance, conditional canonical-action disagreement, joint disagreement exposure, missed-risk rate among accepted actions, drift detection, delay, trace-level false alarm, fit time, amortized batched inference, and serialized bytes. All reported 95% intervals resample complete test traces 1,000 times; no row-level independence is assumed. Runtime is restricted to one Open Multi-Processing (OpenMP)/Basic Linear Algebra Subprograms (BLAS) thread.

All generated results carry a status string that distinguishes the full benchmark from the smoke test. Headline tables and numerical macros are generated from comma-separated outputs; a fail-closed claim audit checks the remaining prose values against those same files. The data manifest stores SHA-256 hashes and byte counts for every selected upstream file. Table VI fixes the split, seeds, calibration level, model capacity, and execution environment used to regenerate the reported evidence.

TABLE VI
FIXED DATA SPLIT, CALIBRATION, MODEL, AND EXECUTION CONTROLS USED FOR REPRODUCIBILITY.

Item	Fixed value
Upstream revision	`bd86629d07d5`
Global seed	20,260,712
Mapper fit region	first 60% of `exp1`
Paired anchors	10% of fit region
Calibration	last 40%, stride four
Conformal level	$\alpha = 0.05$ unless swept
Residual model	70 trees/KPM, depth five
Drift anchors	every 10 samples; $W = 5$
Bootstrap	1,000 whole-trace resamples
Threads	one OpenMP/BLAS thread

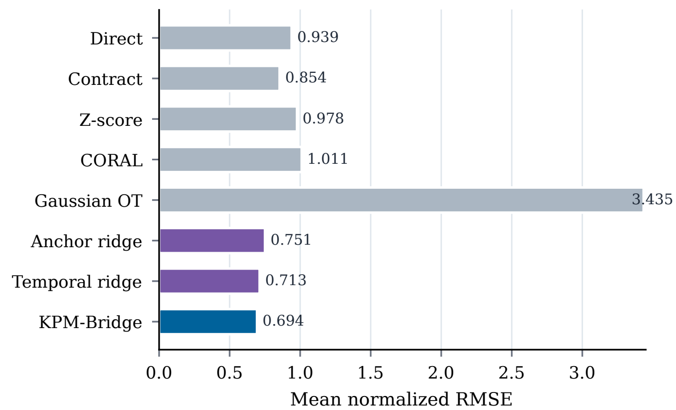

Fig. 4. Stationary normalized reconstruction error across profiles P1–P3 for all portable methods. Lower values indicate more accurate canonical KPM recovery.

VII. RESULTS

A. Stationary Cross-Implementation Portability

Figs. 4 and 5, together with Table VII, summarize P1–P3. KPM-Bridge obtains mean NRMSE 0.694, an observed 2.6% reduction relative to temporal ridge, the strongest reconstruction baseline. It reaches 89.34% decision agreement, 0.53 percentage points above the same strongest decision baseline. Its mean label regret is 0.0977, compared with 0.1033 for CORAL and 0.1019 for temporal ridge. Because these margins are small relative to the whole-trace bootstrap interval widths reported below, we treat the stationary comparison as accuracy parity with the strongest baselines rather than as a significance claim.

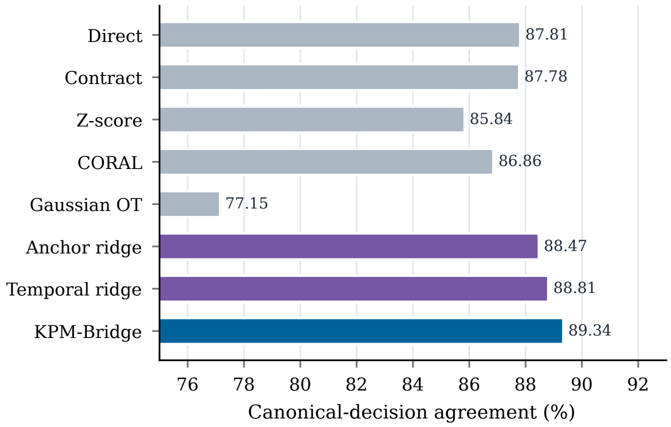

Fig. 5. Fixed-xApp decision agreement across stationary profiles P1–P3. Higher values indicate closer agreement with decisions made from canonical inputs.

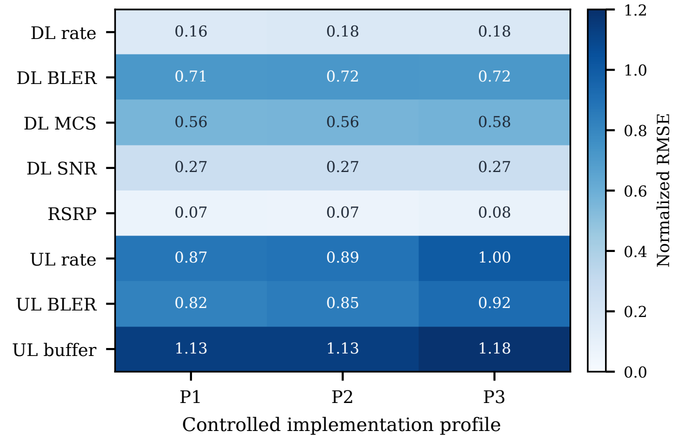

Fig. 6. Per-KPM normalized reconstruction error for KPM-Bridge across profiles P1–P3, exposing feature-level variation hidden by the pooled metric.

TABLE VII
MEAN STATIONARY RECONSTRUCTION, DECISION, REGRET, AND RUNTIME PERFORMANCE ACROSS PROFILES P1–P3.

Method	NRMSE↓	Agree (%)↑	Regret↓	μs/sample
Direct	0.939	87.81	0.1097	0.03
Contract	0.854	87.78	0.1040	1.02
Z-score	0.978	85.84	0.1025	0.04
CORAL	1.011	86.86	0.1033	0.04
Gaussian OT	3.435	77.15	0.1249	0.04
Anchor ridge	0.751	88.47	0.1034	1.31
Temporal ridge	0.713	88.81	0.1019	2.34
KPM-Bridge	**0.694**	**89.34**	**0.0977**	**10.39**

What distinguishes KPM-Bridge at this matched accuracy is the accompanying evidence: it alone emits per-sample meaning, age, support, and calibration state, and the selective results below show that this evidence, not raw accuracy, is what improves the quality of admitted decisions.

Per-profile KPM-Bridge NRMSE is 0.675, 0.682, and 0.725 for P1–P3. Whole-trace bootstrap 95% intervals are $[0.600, 0.749]$, $[0.607, 0.759]$, and $[0.636, 0.817]$. The intervals expose substantial scenario heterogeneity that a row-level interval would suppress. Decision agreement ranges from 87.94% to 90.13%.

The counter-only contract baseline degrades in P2 because a missing adjacent report makes that local rate unavailable, while counter noise and quantization are amplified by differentiation. Anchor ridge reduces that error, whereas the temporal models use history and missing indicators. Gaussian OT also degrades because matching the marginal distribution of a cumulative counter to a rate does not reconstruct eventwise increments. These cases illustrate why distribution alignment cannot replace counter semantics.

The direct baseline retains 87.81% agreement despite poor reconstruction because the cost-sensitive xApp intervenes on 88.66% of canonical test samples. This decision imbalance makes agreement easier and is why NRMSE, regret, and selective disagreement are reported together.

B. Featurewise Error Structure

Figs. 6 and 7 prevent pooled NRMSE from hiding difficult features. Downlink-rate NRMSE is 0.162–0.176, and reference signal received power remains below 0.080. Uplink buffer is hardest at 1.13–1.18 because it is zero-inflated with rare bursts; uplink rate and error rate are also more variable than their downlink counterparts. Relative to temporal ridge, the nonlinear residual reduces mean downlink-rate error by 22.8% and reference-power error by 42.6%, but changes downlink-BLER NRMSE by only 0.108%. The benefit is therefore feature dependent: the residual helps nonlinear scaling and saturation, whereas sparse burst targets remain anchor limited.

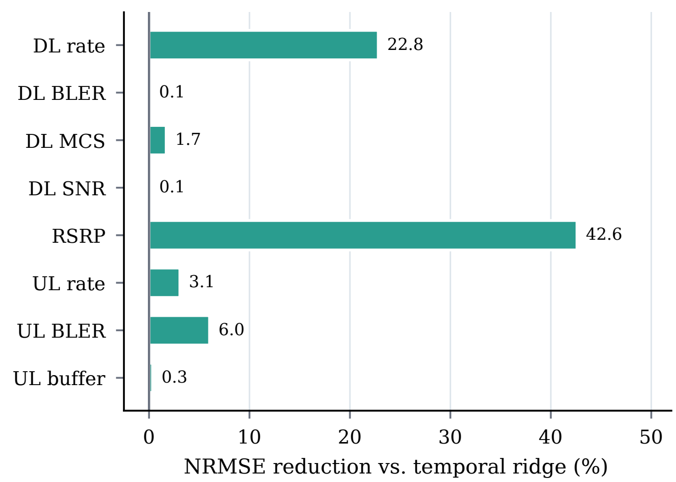

Fig. 7. Featurewise NRMSE reduction relative to temporal ridge. Positive values indicate lower reconstruction error for KPM-Bridge.

The featurewise result also explains why decision agreement improves less than NRMSE. The risk xApp is driven mainly by downlink rate and downlink block error rate. KPM-Bridge substantially improves the former but not the latter; improvements to reference power and uplink metrics affect the logistic margin only indirectly. A different xApp would induce a different functional certificate and may value the same reconstruction errors differently.

C. Calibration and Selective Inference

Table VIII shows that, across P1–P3 at $\alpha = 0.05$, empirical joint coverage averages 94.32%, 0.68 percentage points below the nominal 95%. The experiment-level split separates repetitions, but temporal dependence means that exchangeability

TABLE VIII
VALID-REGIME JOINT COVERAGE, WHOLE-TRACE ACCEPTANCE AND CONDITIONAL CANONICAL-ACTION DISAGREEMENT, AND DRIFT DIAGNOSTICS AT $\alpha = 0.05$.

Profile	Valid cov. (%)	Accept (%)	Cond. dis. (%)	Cluster hi.	Detect (%)	Delay (s)
P1	94.06	57.91	0.228	0.403	–	–
P2	94.45	56.78	0.202	0.361	–	–
P3	94.45	48.66	0.198	0.352	–	–
P4	94.43	35.24	6.030	8.181	88.7	8.96

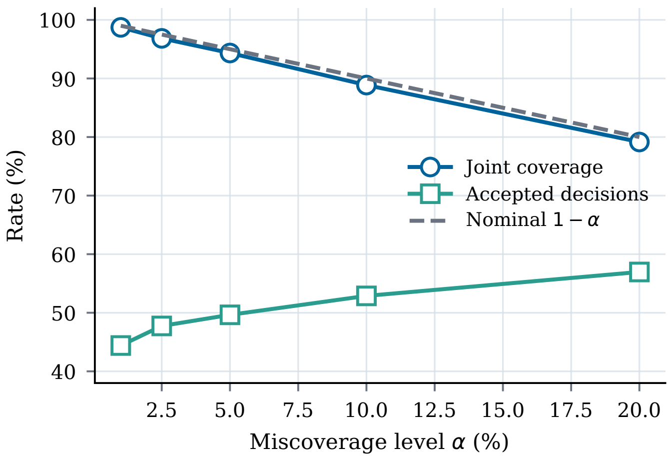

Fig. 8. Empirical joint coverage and accepted-decision rate versus conformal miscoverage level α. The dashed line is the nominal coverage target $1 - \alpha$.

remains an approximation. The average generic radius $\hat{q}_\alpha$ in (8) is 3.324 training scales.

The functional interval I_t^ℓ is more selective than the generic ball for the registered logit in (16). KPM-Bridge accepts 54.45% of P1–P3 actions with mean conditional canonical-action disagreement 0.209%. Per-profile whole-trace upper confidence limits are 0.403%, 0.361%, and 0.352%; no zero-error claim is made. Removing $\hat{q}_\alpha^\ell$ raises disagreement to 9.67–11.73% while accepting 78.10–92.18% of samples.

Figs. 8 and 9 sweep the order statistics $\hat{q}_\alpha$ and $\hat{q}_\alpha^\ell$. Mean coverage across P1–P4 decreases from 98.72% at $\alpha = 0.01$ to 79.15% at $\alpha = 0.20$, while acceptance rises from 44.41% to 56.96%. Conditional disagreement rises from 1.51% to 1.94%. These coupled curves define an operating tradeoff, not a universally optimal α.

D. Drift Invalidation

Before the injected P4 change, coverage is 94.43%. At the 1.25-scale shift, the anchor-residual monitor detects 88.7% of traces with 8.96 s median delay and 9.4% pre-change false alarms at trace level. Three denominators must not be conflated. The full gate accepts 5.99% of post-shift samples; 76.23% of those admitted actions disagree with the fixed xApp on canonical input, so the joint disagreement exposure is 4.57%. Ignoring drift accepts 98.19%, with 87.6% conditional disagreement and 86.0% joint exposure. Thus invalidation reduces exposure by 94.7% mainly through abstention, not by making the few pre-alarm actions accurate.

Figs. 10 and 11 show that a 0.5-scale shift is difficult: detection is 9.43% and post-shift coverage is 90.64%. Detection reaches 71.70% at one scale and 88.68% at 1.25 scales. The

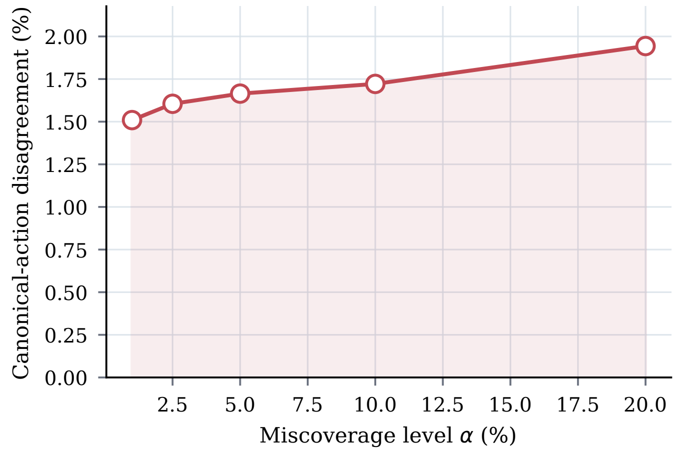

Fig. 9. Mean conditional canonical-action disagreement versus conformal miscoverage α across profiles P1–P4.

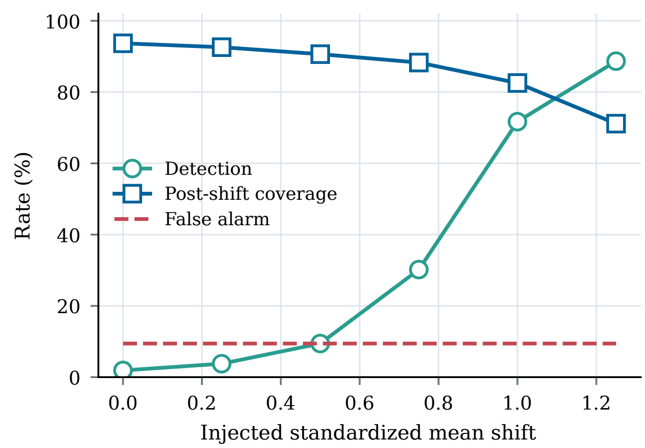

Fig. 10. Detection rate, post-shift coverage, and trace-level false-alarm rate versus the injected standardized mean shift.

false-alarm line remains 9.43% because the pre-change segment is identical throughout the sweep. Removing only the functional uncertainty margin changes conditional disagreement from 76.23% to 76.14% and exposure from 4.57% to 4.73%; under this severe shift, the calibrated interval cannot substitute for prompt epoch invalidation.

E. Ablation and Sensitivity

Table IX shows that contract normalization alone is insufficient (NRMSE 0.854). Current-sample anchor regression reduces NRMSE to 0.751, temporal ridge reaches 0.713, and the full nonlinear residual reaches 0.694. Removing the typed compiler gives 0.695 because the benchmark preserves column order and supplies paired anchors. This negative result is intentional: typing is justified by fail-closed semantics, not an accuracy gain on already paired columns. The executable compiler accepts the declared conversion and permutation checks and rejects both quantity and scope decoys (4/4 checks), a guarantee the untyped variant cannot provide.

Figs. 12, 13, and 14 separate anchor scarcity, unseen missingness, and unseen lag. Increasing P3 anchors from 1% to 5% lowers NRMSE from 0.760 to 0.724; 20% yields 0.721, showing diminishing returns in the objective (4). A P1 mapper

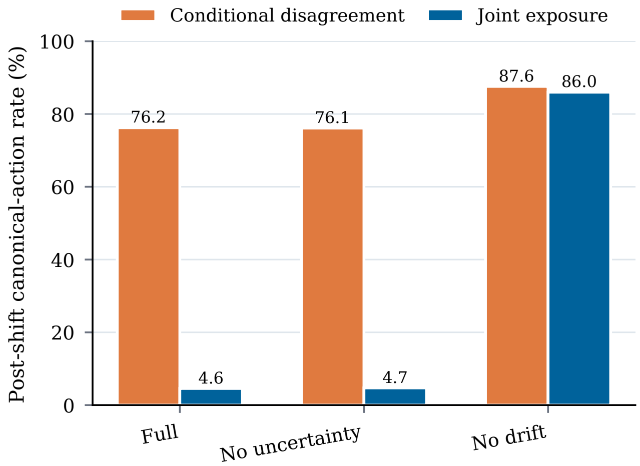

Fig. 11. Post-shift P4 conditional canonical-action disagreement and joint disagreement exposure for the full and ablated gates.

TABLE IX
MEAN STATIONARY ABLATION RESULTS ACROSS P1–P3, WITH THE MECHANISM RETAINED BY EACH VARIANT.

Variant	Retained mechanism	NRMSE	Agree (%)
No residual/temporal	contract normalization only	0.854	87.78
No temporal	current report only	0.751	88.47
Linear residual	ridge instead of trees	0.713	88.81
No typed contract	statistical fit; no type rejection	0.695	89.32
Full	contracts + temporal nonlinear residual	0.694	89.34

trained at 3% loss is evaluated without refitting: agreement declines from 90.15% with no loss to 88.34% at 30% loss as the mask and support terms in (6) deteriorate. The expected one-sample lag gives 90.13% agreement; removing it gives 89.21%, while two or more samples reduce agreement below 87.6%. Six samples also exceed the 1.5-s age budget and are rejected even when the mapper returns a vector.

VIII. Complexity, Overhead, and Deployment

The Python reference ran on an AMD EPYC 9V74 central processing unit (CPU) with one OpenMP/BLAS thread. KPM-Bridge fits a profile in 1.00–1.44 s, takes 10.4 μs/sample in amortized batches, and occupies 1.45 MiB on average; temporal ridge takes 2.34 μs/sample and 13.4 KiB. Fig. 15 and Table X distinguish measured batch cost and storage from asymptotic scaling. They exclude E2 decoding, transport, scheduling, queueing, and actuation and therefore are not worst-case deadline claims.

The source cadence is 250 ms and the QoS horizon is one second. Although measured batch compute is small relative to that cadence, deployment requires a joint tail-latency budget for decoding, transport, scheduling, certificate lookup, inference, and actuation.

At report rate r, the certificate contributes $48rNV$ bytes/s for N UEs and V certified views. For $r = 10$, $N = 100$, and $V = 1$, this is 0.384 Mbit/s; the eight-float32 vector raises the total to 0.640 Mbit/s before headers. The evaluated sidecar stays inside the Near-RT RIC and does not alter E2 subscription payloads; interface transport would add protocol and security headers.

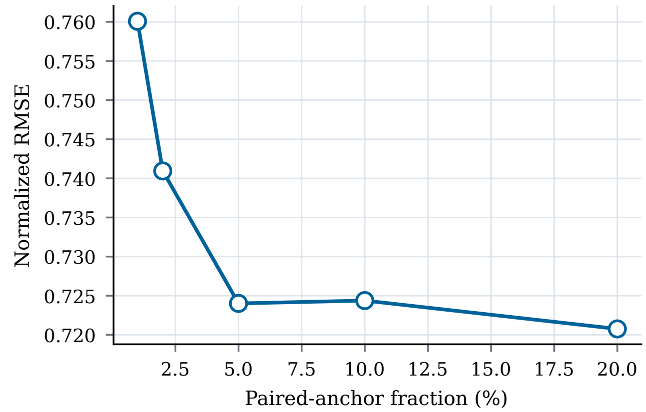

Fig. 12. P3 reconstruction error versus the paired-anchor fraction. The curve shows diminishing returns beyond the initial anchor increase.

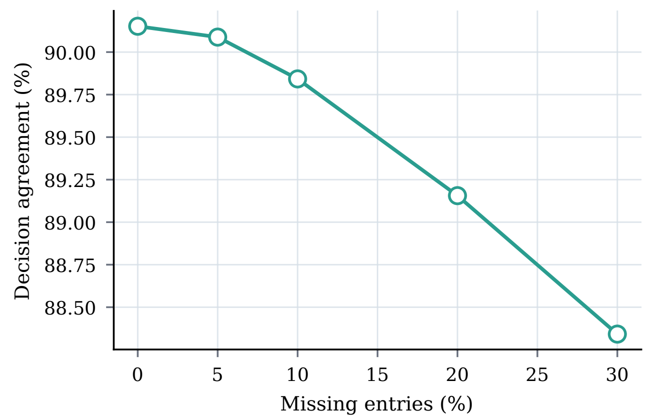

Fig. 13. Fixed-xApp decision agreement when a P1 mapper trained at 3% loss is evaluated under increasing missing-entry rates without refitting.

Compilation occurs only after a schema or policy change. Per-implementation streaming state comprises nine report positions, masks, age/support, model state, and drift statistics. E2 nodes scale linearly after compilation and can be sharded. Production deployment still requires a compact tree runtime and target-RIC tail-latency measurements.

IX. Discussion and Limitations

Certificate interpretation: The certificate binds a mapping, model, calibration epoch, and measured operating evidence. Its finite-sample guarantee requires the stated exchangeability assumption and is neither per-subgroup conditional coverage nor proof that an xApp is beneficial, fair, secure, or RAN-stable: it certifies telemetry fitness, not system trustworthiness.

Information loss and anchors: Cell aggregation cannot identify UEs without side information, long windows suppress transients, and consecutive losses enlarge uncertainty, so the bridge exposes support and abstains. A cybersecurity-driven quantum digital twin similarly maps O-RAN observations into detection registers [40]; KPM-Bridge guards input compatibility, not policy. The 10% anchor budget needs a trusted reference, digital twin, temporary dual reporting, or validated probe; performance remains useful at 5%, but anchor cost and integrity are deployment constraints.

Controlled profiles versus vendors: P1–P4 are controlled profiles over a public experimental corpus, not commercial

TABLE X
ASYMPTOTIC COMPUTATION, PERSISTENT STATE, AND MEASURED REFERENCE COST BY KPM-BRIDGE LIFECYCLE STAGE.

Stage	Trigger	Time complexity	Persistent memory	Reference observation
Contract assignment	schema/policy change	$O((p_v + d)^3)$	$O(p_v d)$ costs	negligible for eight targets
Counter/event-time update	every report	$O(hd)$	$O(hd)$ history	included in mapper timing
Gradient-boosted mapping	every report	$O(dN_{\mathrm{tr}}L_{\mathrm{tr}})$	$O(dN_{\mathrm{tr}}N_{\mathrm{leaf}})$	8.2–14.1 μs/sample batched
Generic/functional gate	every report	$O(d)$	$O(1)$ quantiles	fixed 48-byte certificate
Mean-shift monitor	trusted anchor	$O(d^2 + Wd)$	$O(d^2 + Wd)$	once per 2.5 s in benchmark
Refit/recalibration	invalid epoch	offline/model dependent	calibration buffer	1.00–1.44 s profile fit

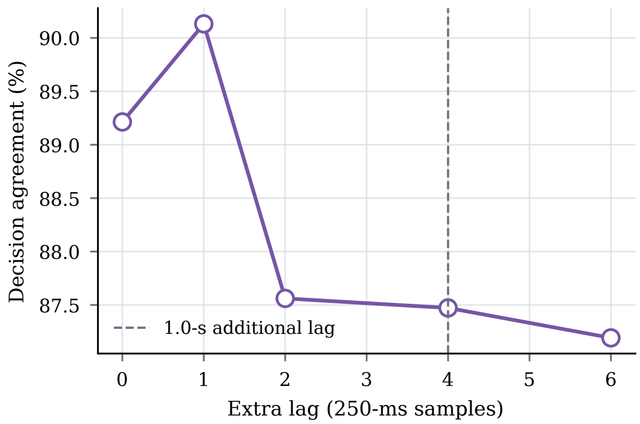

Fig. 14. Fixed-xApp decision agreement under unseen additional lag. The dashed line marks the one-second age increment reached at four 250-ms samples.

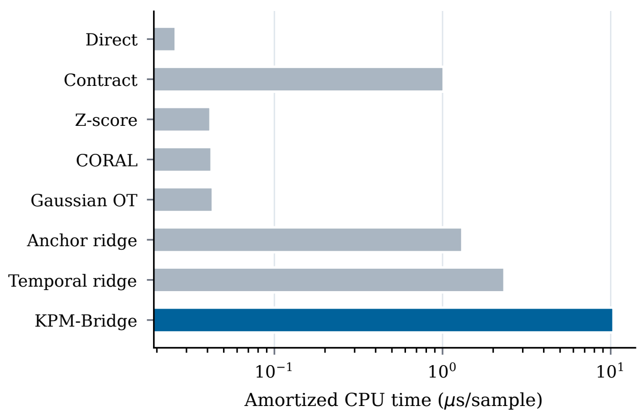

Fig. 15. Amortized single-thread processor inference time per sample for KPM-Bridge and all portable baselines; the horizontal axis is logarithmic.

products. Static UEs and a 250-ms cadence limit generalization to mobility, handovers, multi-cell conversion, and faster loops. Vendor studies should disclose, where permitted, report schemas, timestamp behavior, aggregation/reset tests, and anchor procedures.

Contract governance and versioning: Operator-governed contracts bind quantity/scope namespaces, E2SM revision, measurement identifier, labels, canonical ontology, and transform hash. Units may convert, UE-to-cell scope is breaking, and window or clock changes force recalibration; the operator, not the xApp, approves sources against model-card requirements for features, age, support, and functional score.

Calibration under time dependence: Split-conformal coverage is exact for exchangeable calibration units. Experiment separation and subsampling reduce, but do not remove, dependence; the measured 94.32% coverage at nominal 95% motivates block-aware or weighted calibration [29].

Drift and false alarms: The anchor-residual detector cannot observe drift without reference evidence. Under P4, detection is imperfect, 9.4% of traces trigger early, and post-shift conditional disagreement remains 76.23%; its main benefit is the 94.7% exposure reduction achieved through abstention. Lower thresholds trade delay for availability, and per-stream-maximum calibration is empirical, not an anytime-valid false-alarm theorem, so alarms need cause codes and operator diagnostics. An adversary corrupting reports and anchors can evade residual monitoring, and induced missingness can force denial of service; authenticated anchors, access or execution controls, rate limits, redundant evidence, and emergency policies remain platform responsibilities.

External validity: Validation should combine xDevSM and KPM-Bridge on OAI, srsRAN, and vendor E2 traces while varying granularity and schedulers. Conditional or block-aware calibration, monotone maps, and cost-aware anchor selection also remain open.

X. REPRODUCIBILITY

Source, pinned manifests, tests, and generated artifacts are available in the KPM-Bridge repository. One command regenerates the reported evidence; upstream GPL-3.0 traces are not redistributed.

XI. CONCLUSION

KPM-Bridge separates E2/API interoperability from the semantic and statistical evidence required by a portable AI xApp. Typed contracts prevent quantity and scope substitution; event-time mapping repairs declared counter and temporal differences; conformal sets expose uncertainty; and drift monitoring invalidates stale calibration. On independent ColO-RAN repetitions, the method matches the strongest evaluated portable baselines on stationary accuracy while making abstention explicit; empirically, the functional gate yields 0.209% conditional canonical-action disagreement at 54.45% acceptance. Under severe drift, fail-closed gating reduces joint disagreement exposure by 94.7%, but only by accepting 5.99% of post-shift samples; conditional disagreement, detection delay, and false alarms remain material limitations. Portable xApps should receive both KPM values and verifiable evidence of their meaning, age, support, and calibration regime.

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